

2.a Discuss the importance of Analytics in managing a Vaccine Supply Chain effectively and safely

**Introduction**

A vaccine supply chain is the process of manufacturing, transporting, storing and distributing vaccines until they reach patients. Since vaccines are temperature-sensitive, analytics plays an important role in maintaining their quality, safety and effectiveness.

**Cold Chain:** The cold chain is a temperature-controlled supply chain used for vaccines and pharmaceutical drugs. It consists of: Transport and storage equipment, Trained personnel, Efficient management procedures

Most vaccines are generally maintained at 2–8°C. Improper temperature can reduce vaccine potency.

**Importance of Analytics**

1. Real-Time Monitoring: Sensors continuously collect temperature and humidity data.

Storage conditions can be monitored in real time.

Location-tagged data helps identify where a problem occurs.

2. Preventing Vaccine Spoilage: Analytics detects abnormal temperature conditions.

It can identify overheating, freezing and improper storage.

Early warnings help prevent vaccine wastage.

3. Descriptive Analytics: It answers "What is happening?"

Shows temperature trends and storage conditions.

Provides reports and dashboards.

Helps personnel understand the current condition of the cold chain.

4. Predictive Analytics: It answers "What is likely to happen?"

Predicts possible temperature violations.

Detects possible compressor or power failure.

Estimates how long the storage unit can remain safe.

Gives warnings before vaccines are spoiled.

5. Prescriptive Analytics: It answers "What should be done?"

Suggests the optimal temperature setting.

Helps balance freezing and spoilage risks.

Recommends how vaccines should be distributed among storage units.

Supports equipment purchase and maintenance decisions.

**Magpie Sensing Example:**

The PDF describes Magpie Sensing, which provides wireless temperature and humidity monitoring for cold-chain products.

It uses:

- Real-time temperature monitoring
- Location-aware tracking
- Web dashboards
- Analytics algorithms
- Alerts and warnings

Magpie Sensing applies descriptive, predictive and prescriptive analytics to convert raw monitoring data into useful recommendations.

**Benefits:** Analytics helps to:

- Reduce vaccine spoilage and wastage.
- Improve cold-chain efficiency.
- Detect human and equipment errors.
- Support quick decision-making.
- Reduce unnecessary costs.
- Maintain safety and regulatory compliance.
- Ensure effective vaccines reach patients.

Analytics makes the vaccine supply chain safer, more efficient and reliable. By using descriptive, predictive and prescriptive analytics, organizations can monitor vaccine conditions, predict problems and take corrective actions before vaccines are damaged.

### 5.b Write notes on Text Mining Tools and NLP

**A. Text Mining Tools**

**Definition:** Text mining tools are software applications used to process, analyze and extract useful information from textual data.

The PDF mainly discusses commercial tools and free/open-source tools.

**Commercial Tools**

1. ClearForest: Provides text analysis and visualization.

Helps extract useful information from text.

2. IBM SPSS Modeler: Provides data and text analytics capabilities.

Used for analytical processing and knowledge discovery.

3. SAS Text Miner: Provides text-processing and text-mining capabilities.

4. Megaputer Text Analyst: Supports: Semantic analysis, Summarization, Clustering, Natural language retrieval

5. WordStat :Used for analyzing interviews, questionnaires and other textual data.

**Open-Source Tools**

1. RapidMiner

Provides data-mining and text-mining capabilities.

Uses a graphical drag-and-drop interface.

2. GATE

Open-source toolkit for text mining.

Provides a framework and graphical development environment.

3. LingPipe

Java libraries for linguistic analysis of human language.

4. Open Calais

Provides semantic analysis capabilities for websites and applications.

These tools help convert textual information into useful structured information and knowledge.

**B. Natural Language Processing (NLP)**

**Definition:** Natural Language Processing (NLP) is a subfield of Artificial Intelligence and Computational Linguistics that deals with processing and understanding human language.

Its goal is to convert human language into a form that computers can process.

**Role of NLP in Text Mining:**

NLP helps to:

- Process unstructured text.
- Extract important features.
- Identify words and their relationships.
- Understand grammatical structures.
- Convert text into structured information.

Therefore: Text Mining = NLP + Data Mining

**Major NLP Tasks**

1. Part-of-Speech Tagging: Identifies whether a word is a: Noun, Verb, Adjective, Adverb

2. Text Segmentation: Divides text into meaningful units such as words or sentences.

3. Word Sense Disambiguation: Determines the correct meaning of a word based on context.

Example: "Bank" may mean a financial institution or river bank.

4. Question Answering: Automatically answers questions written in natural language.

5. Automatic Summarization: Creates a shorter version of a document containing important points.

6. Machine Translation: Automatically translates one human language into another.

7. Speech Recognition: Converts spoken language into machine-readable text.

8. Text-to-Speech: Converts written text into spoken language.

9. Optical Character Recognition: Converts printed or handwritten text from images into editable text.

**Conclusion:** Text Mining Tools provide the software needed to analyze large amounts of textual data, while NLP provides the language-processing techniques needed to understand that data. Together, they help organizations extract useful information, patterns and knowledge from unstructured text.

### 3.b Write notes on Business Analytics and Big Data Analytics

**Business Analytics**

**Definition**

Business Analytics is the process of developing actionable decisions or recommendations based on insights obtained from data. It combines computer technology, statistics and management science techniques to solve real-world problems.

**Types of Business Analytics**

1. Descriptive Analytics

Answers "What is happening?"

Uses reports, dashboards, queries and visualization.

Helps understand historical and current business performance.

Example: Monthly sales report showing a decrease in sales.

2. Predictive Analytics

Answers "What is likely to happen?"

Uses statistics and data-mining techniques.

Helps predict customer behavior, sales and risks.

Example: Predicting whether a customer will leave a company.

3. Prescriptive Analytics

Answers "What should we do?"

Recommends the best possible action.

Uses optimization and management science techniques.

Example: Finding the best price or production plan.

**Big Data Analytics**

**Definition:** Big Data refers to data that is too large or complex to be stored and processed using a single storage unit. It may be structured, unstructured or streaming data.

**Characteristics of Big Data**

1. Volume: Refers to the huge quantity of data generated.
2. Velocity: Refers to the high speed at which data is generated and processed.
3. Variety: Refers to different forms of data such as: Text, Images, Videos, Social media data, Sensor data, Big Data Technologies

**HDFS:** Hadoop Distributed File System (HDFS) stores large data by dividing it into chunks and distributing them across multiple machines.

**MapReduce:** MapReduce processes Big Data by moving computation closer to where the data is stored, allowing large datasets to be processed efficiently.

Examples: Google: Processes and indexes billions of Web pages.

Algorithmic trading: Processes financial data at very high speed.

Sentiment analysis: Analyzes social media and customer responses.

**Conclusion**

Business Analytics helps organizations understand data and make better decisions, while Big Data Analytics enables organizations to analyze very large, fast and diverse datasets. Both are important for modern data-driven decision-making.

1. Challenges involved in developing a Predictive Model and deploying & maintaining it in a real-world environment

**Introduction:** Predictive modelling uses historical data and machine learning algorithms to predict future outcomes. The challenges are different during model development and during deployment and maintenance.

**A. Challenges in Model Development**

1. Problem Definition: The business problem must be clearly defined.

The correct target variable should be identified.

A wrong problem definition can lead to an incorrect model.

2. Data Collection: Relevant data must be collected from suitable sources.

Data may contain missing, duplicate or incorrect values.

Insufficient data can reduce model quality.

3. Data Preparation: Missing values must be handled.

Outliers and errors must be identified.

Data may need transformation and normalization.

Good-quality data is essential for accurate predictions.

4. Feature Selection: Important variables must be selected.

Irrelevant features can reduce model performance.

New useful features may also be created.

5. Model Selection and Training: A suitable algorithm must be selected.

Examples include decision trees, logistic regression and neural networks.

The model is trained using historical data.

6. Model Evaluation

The model must be tested using unseen data.

Measures such as accuracy, precision, recall, RMSE and R<sup>2</sup> can be used.

Overfitting must be avoided.

**B. Challenges in Deployment**

1. System Integration

The trained model must be integrated with applications, databases or APIs.

It should work correctly with real-time input.

2. Scalability

The system should handle increasing users and data.

Prediction speed should remain acceptable.

3. Data Changes

Real-world data may change over time.

New patterns may differ from training data.

- C. Challenges in Maintenance

1. Model Performance Monitoring

Accuracy must be continuously monitored.

A decrease in performance should be detected.

2. Data Drift

Input data may change over time.

The model may need retraining with new data.

3. Updating the Model

New records should be incorporated periodically.

The model should be retrained when necessary.

**Conclusion**

Model development mainly focuses on data, features, algorithms and evaluation, whereas deployment and maintenance focus on integration, performance, data changes and continuous updating. A predictive model is successful only when it remains useful in the real-world environment.

### 4.a What are Analytics and Text Mining? Explain with examples.

**Text Analytics**

**Definition:** Text Analytics is the process of extracting valuable information, patterns and insights from unstructured textual data using computational techniques and algorithms.

**Main Objectives**

1. Information Extraction

Extracts important information such as: Names, Dates, Locations, Organizations

2. Sentiment Analysis

Identifies the emotional tone of text as: Positive, Negative, Neutral

Example: Analyzing customer reviews to measure satisfaction.

3. Text Classification

Places documents into predefined categories.

Example: Classifying e-mails as spam or non-spam.

4. Topic Modeling: Identifies major topics or themes in a collection of documents.

5. Text Summarization: Creates a shorter version of a lengthy document containing important information.

6. Pattern Recognition: Finds repeated patterns and trends in textual data.

**Text Mining**

**Definition:**Text Mining is the semi-automated process of extracting useful patterns, information and knowledge from large amounts of unstructured text.

Unlike traditional data mining, which mainly works with structured databases, text mining works with: PDF files, Word documents, E-mails, Web pages, XML files

**Text Mining Process**

1. Structure the Text: Unstructured text is converted into a suitable structured form using: Tokenization, Stop-word removal, Stemming, Part-of-speech tagging
2. Extract Knowledge: Data-mining techniques are then applied, such as: Classification, Clustering, Pattern discovery

**Information extraction**

**Important Techniques**

Tokenization: Splitting text into words or tokens.

Stop-word removal: Removing common words such as "the", "and", etc.

Stemming: Reducing words to their root form.

TF-IDF: Measures the importance of a word in a document.

NER: Identifies people, places and organizations.

**Examples of Text Mining**

Marketing: Analyzing customer reviews, complaints and social media comments.

E-mail: Filtering spam and prioritizing important messages.

Healthcare: Finding relationships between diseases, genes and proteins from medical literature.

Academic Research: Analyzing research papers to discover important topics and trends.

**Difference**

**Text Analytics** **Text Mining**

**Broader concept**      **Specific process within text analytics**

**Extracts insights from text**      **Discovers patterns and knowledge**

**Includes retrieval, extraction and analysis**      **Uses data-mining techniques on text**

**Conclusion:** Text Analytics and Text Mining help organizations convert unstructured text into useful information and knowledge. They are widely used in marketing, healthcare, e-mail processing, research and many other fields.

2. Explain different types of Analytics – Descriptive, Diagnostic, Predictive and Prescriptive with examples

**Introduction**

Analytics is the process of examining data to obtain useful insights and support decision-making. The four major types are Descriptive, Diagnostic, Predictive and Prescriptive Analytics.

1. Descriptive Analytics: Descriptive analytics answers: "What happened?"

It analyzes historical data and summarizes past performance.

Uses: Reports, Dashboards, Charts, Trends, Historical analysis

Example: A supermarket analyzes last year's sales to find which products were sold the most.

2. Diagnostic Analytics: Diagnostic analytics answers: "Why did it happen?"

It investigates the causes behind a particular event or result.

Uses: Finding relationships, Identifying causes, Comparing different factors, Understanding patterns

Example: If sales of a product decreased, diagnostic analytics can identify whether the reason was poor quality, high price or low demand.

3. Predictive Analytics: Predictive analytics answers: "What is likely to happen?"

It uses historical data, statistics and machine learning to forecast future outcomes.

Uses: Customer churn prediction, Sales forecasting, Loan default prediction, Disease prediction, Demand forecasting

Example: Amazon recommends products based on a customer's previous purchases. Netflix recommends movies based on previous viewing behavior.

4. Prescriptive Analytics: Prescriptive analytics answers: "What should we do?"

It recommends the best action based on available data and predictions.

Uses: Optimization, Resource allocation, Pricing decisions, Production planning, Automated decision-making

Example: A company uses predicted demand to decide how much stock should be ordered.

**Simple Comparison**

| Type         | Main Question      | Example                   |
|--------------|--------------------|---------------------------|
| Descriptive  | What happened?     | Sales report              |
| Diagnostic   | Why did it happen? | Reason for sales decrease |
| Predictive   | What will happen?  | Future sales prediction   |
| Prescriptive | What should we do? | Best stock/order decision |

**Conclusion:** The four types of analytics move from understanding the past to deciding the best future action: Descriptive → Diagnostic → Predictive → Prescriptive

They help organizations make data-driven and effective decisions.

3. A bank develops a Predictive Model to identify customers likely to default on loans. Which measure of success should be given more priority? Justify.

**Introduction**  
A bank wants to predict whether customers are likely to default on loans. This is a classification problem, because the result can be:  
Default → Yes  
Default → No  
For this problem, selecting the correct evaluation measure is very important.

**Important Measures**

1. Accuracy  
Accuracy measures the percentage of total predictions that are correct.  
 $\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$   
However, accuracy alone may not be suitable.

2. Precision  
Precision answers:  
"Among customers predicted as defaulters, how many are actually defaulters?"  
 $\text{Precision} = \frac{TP}{TP + FP}$   
High precision reduces the number of good customers incorrectly classified as defaulters.

3. Recall  
Recall answers:  
"Among all actual defaulters, how many did the model identify?"  
 $\text{Recall} = \frac{TP}{TP + FN}$   
For loan-default prediction, identifying actual risky customers is very important.

4. F1 Score  
F1 Score provides a balance between precision and recall.  
 $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$   
Measure to Give Priority  
Recall should be given high priority.  
The main objective of the bank is to identify as many actual loan defaulters as possible.  
If recall is low:  
Many risky customers will be missed.  
The bank may approve loans to customers who later default.  
This can result in financial losses.  
However, precision and F1-score should also be monitored, because predicting too many good customers as defaulters can unnecessarily reject genuine borrowers.  
The PDF specifically recommends Precision, Recall and F1 for imbalanced data such as fraud detection.

**Example**  
Suppose there are 1,000 customers but only 50 are likely to default.  
A model could predict almost everyone as "No Default" and still achieve high accuracy. But it would fail to identify the actual defaulters.  
Therefore, accuracy alone is misleading.

**Final Choice**  
Priority → Recall  
Also consider → Precision and F1 Score  
This combination gives the bank a better understanding of how effectively the model identifies risky borrowers.  
Conclusion : For loan-default prediction, Recall should be prioritized, because missing an actual defaulter can cause significant financial loss. Precision and F1-score should also be considered to maintain a balance between identifying risky customers and avoiding unnecessary rejection of genuine customers.

6. Discuss Decision Trees, Logistic Regression, KNN, Neural Networks and Bayesian Methods based on working principle, advantages, limitations and applications.

**Introduction:** Predictive analytics uses different machine learning methods depending on the problem. Important methods are Decision Trees, Logistic Regression, KNN, Neural Networks and Bayesian Methods.

1. Decision Tree: A Decision Tree is a flowchart-like model where internal nodes represent conditions, branches represent outcomes and leaf nodes give the final prediction. It produces simple if-then rules.  
Advantages: Easy to understand and interpret; handles numerical and categorical data.  
Limitations: Can overfit when the tree is too deep; small data changes may produce a different tree.  
Applications: Loan approval, medical diagnosis, customer classification.

2. Logistic Regression: It is mainly used for binary classification. It predicts the probability of an outcome using the sigmoid function, which gives a value between 0 and 1.  
Advantages: Simple to implement; provides probabilities; works well for binary outcomes.  
Limitations: Assumes a linear relationship between inputs and log-odds; not suitable for highly complex relationships.  
Applications: Credit scoring, loan approval, customer churn and medical diagnosis.

3. K-Nearest Neighbors (KNN): KNN predicts a new data point using its K nearest training examples. Distance is commonly calculated using Euclidean distance. For classification, the majority class is selected.  
Advantages: Simple to understand; requires little training; useful for classification and regression.  
Limitations: Slow for large datasets; sensitive to feature scaling; performance depends on the value of K.  
Applications: Pattern recognition, customer classification, recommendation systems and image classification.

4. Neural Networks: Neural networks consist of input, hidden and output layers. Each neuron calculates a weighted sum, adds bias and applies an activation function. During training, weights are updated using backpropagation.  
Advantages: Learns complex non-linear patterns; works well with large datasets.  
Limitations: Requires large data; training can be slow; difficult to interpret.  
Applications: Fraud detection, medical diagnosis, customer churn and credit scoring.

5. Bayesian Method: Bayesian methods use probability and Bayes' theorem to update the probability of an event when new evidence is available. The basic idea is: Prior Probability + Evidence → Posterior Probability.  
Advantages: Handles uncertainty well; incorporates prior knowledge; useful for probability-based decisions.  
Limitations: Depends on probability assumptions; complex models may require high computation; incorrect prior assumptions can affect results.  
Applications: Spam detection, medical diagnosis and risk assessment.  
Conclusion: Each method has different strengths. Decision Trees and Logistic Regression are easy to interpret, KNN is simple, Neural Networks handle complex patterns, and Bayesian methods are useful for uncertainty-based decisions.

3. Explain the various processing steps involved in setting up a Predictive Analytics problem

**Introduction:** Setting up a predictive analytics problem involves converting a real-world problem into a form that can be solved using data and predictive models.

1. Understand the Problem: Clearly identify what needs to be predicted.  
Identify the given information, unknown information and goal.  
Define the business objective.  
Example: Predict customer churn or future sales.  
This is the most important step because misunderstanding the problem can make the entire solution incorrect.

2. Gather Inputs / Data: Collect relevant data from suitable sources.  
Sources may include:: Databases, Data warehouses, APIs, Logs, Surveys, Sensors  
Identify assumptions, conditions and constraints.

3. Process / Prepare the Data: Raw data is rarely ready for modelling.  
It may require: Removing duplicate data.  
Handling missing values.  
Correcting errors.  
Handling outliers.  
Transforming data.  
Scaling or normalizing values.  
Good-quality data improves model performance.

4. Feature Selection and Engineering: Select important variables that influence the prediction.  
Remove irrelevant features.  
Create new useful features from existing variables.  
Example: Income – Expenses = Savings

5. Model Selection  
Choose an appropriate predictive technique based on the problem.  
Examples:  
Regression → Predict numerical values.  
Classification → Predict categories.  
Decision Tree → Rule-based prediction.  
Neural Network → Complex patterns.  
Logistic Regression → Yes/No outcomes.

6. Model Training  
Training data is provided to the selected algorithm.  
The model learns patterns and relationships from the data.

7. Model Evaluation  
Test the model using unseen data.  
Evaluate its accuracy and reliability.  
Metrics include: Accuracy, Precision, Recall, RMSE, R².  
8. Deployment: Put the trained model into a real-world system.  
It can be integrated into: Websites, Applications, APIs  
Dashboards

9. Monitoring and Updating  
Continuously monitor model performance.  
Check for accuracy reduction and data drift.  
Retrain or update the model when required.  
Conclusion

The major processing flow is:  
Problem Definition → Data Collection → Data Preparation → Feature Engineering → Model Selection → Training → Evaluation → Deployment → Monitoring

Following these steps helps create a reliable predictive analytics solution.

1. Finding the Structure of Words – Explain about Morphological Models

**Introduction:** Morphology is the study of the internal structure and formation of words. It explains how words are formed from smaller meaningful units called morphemes.

For example: unhappiness = un + happy + ness  
Here, each part contributes to the meaning of the complete word.  
What is a Morphological Model?  
A morphological model is a method used to represent and analyze how words are formed and how their forms change.

It helps NLP systems to:  
Identify the structure of words.  
Find the root or stem of a word.  
Identify prefixes and suffixes.  
Understand grammatical information.  
Generate different forms of a word.

**Major Morphological Models**

1. Item-and-Arrangement Model  
A word is considered as a combination of smaller units.  
These units are arranged together to form a word.  
The units may be roots, prefixes or suffixes.  
Example: un + happy + ness  
un → prefix  
happy → root  
ness → suffix

2. Item-and-Process Model:  
A basic word is transformed using a morphological process.  
The process may change the form of the word.  
It is useful for representing grammatical changes.  
Example: walk → walked  
The suffix -ed represents past tense.

3. Word-and-Paradigm Model: Words are grouped according to their grammatical forms.  
A word can have different forms depending on tense, number, person, etc.  
Example: go → goes → went → going  
These forms belong to the same word paradigm.

4. Finite-State Morphological Model  
Morphology can be represented using finite-state machines/transducers.  
It maps an input word to its morphological analysis.  
It can also generate word forms from a root and grammatical information.  
Example: cats → cat + plural  
Applications: Morphological models are useful in:  
Spell checking, Machine translation  
Search engines  
Speech processing  
Information retrieval  
Text classification  
Natural Language Processing

**Conclusion:** Morphological models help computers understand the internal structure and formation of words.  
They identify roots, affixes and grammatical information and are therefore an important part of NLP.

5. What is Data Effect? Discuss how quality characteristics of data can influence Predictive model performance

**Introduction:** The performance of a predictive model depends heavily on the quality of the data used for training. Poor-quality data can produce incorrect patterns and unreliable predictions.

**Important Data Quality Characteristics**

1. Accuracy: Data should correctly represent the real-world situation.  
Incorrect values can lead to incorrect predictions.  
Example: Incorrect customer income may result in wrong credit-risk prediction.

2. Completeness: Data should contain the required values.  
Missing values can reduce model accuracy.  
Important features should not be missing frequently.

3. Consistency: Data should be consistent across different sources.  
Different formats or conflicting values can confuse the model.  
Example: One system stores gender as "Male", while another uses "M".

4. Relevance: Only useful data should be included.  
Irrelevant variables may add noise.  
Proper feature selection improves the model.

5. Timeliness: Data should be sufficiently current.  
Old data may not represent current customer behavior or market conditions.  
This is especially important for real-time prediction.

6. Proper Data Distribution: Data should represent the actual population.  
If one class is much larger than another, the model may become biased.  
Example: If 95% of transactions are non-fraudulent, a model may struggle to identify fraud.

**Effect of Poor Data Quality**

Missing Data  
Can reduce the amount of usable training data.  
May produce unreliable predictions.

Incorrect Data  
Creates wrong relationships.  
Leads to inaccurate model results.

Outliers  
Extreme values can influence some models strongly.  
They should be investigated and handled appropriately.

Biased Data  
If training data contains bias, the model may produce biased or unfair predictions.  
The PDF identifies bias as an important predictive analytics challenge.

Duplicate Data  
Repeated records can give some observations excessive importance.  
This may distort the learning process.

Poor Feature Quality  
Irrelevant or weak features may reduce prediction performance.  
Good feature selection is therefore important.

Improving Data Quality  
Data quality can be improved by: Data cleaning, Handling missing values, Removing duplicates, Detecting outliers, Correcting errors, Transforming data, Selecting relevant features, Regularly updating data

**Conclusion**  
Good data → Better model performance  
Poor data → Poor predictions  
Therefore, data quality is one of the most important factors affecting the accuracy, reliability and fairness of a predictive model.

2. Explain Different Components of Words

**Introduction:**  
A word may consist of one or more smaller meaningful units called morphemes. Morphological analysis identifies these components and determines their contribution to the meaning of a word.

1. Root: The root is the main part of a word that carries its basic meaning.  
Examples: teach in teacher, kind in kindness, play in replay  
The root generally forms the central meaning of the word.

2. Stem: A stem is the form to which an affix can be attached.  
For example: unhappiness  
happy → base/root  
unhappy → stem  
unhappiness → complete word

The concept of stemming is also important in text processing; stemming reduces different word forms to their stem or base form.

3. Affixes: Affixes are units attached to a root or stem.  
Prefix: A prefix occurs before the root.  
Examples:  
un + happy → unhappy  
re + write → rewrite  
dis + agree → disagree  
Suffix: A suffix occurs after the root.  
Examples:  
teach + er → teacher  
kind + ness → kindness  
walk + ed → walked

4. Inflectional Morphemes: Inflection changes the grammatical form of a word without creating a completely new word.  
Examples:  
cat → cats  
walk → walked  
play → playing  
They may indicate: Number, Tense, Person, Comparison

5. Derivational Morphemes: Derivational morphemes create a new word or change its meaning/category.  
Examples:  
happy → unhappy  
teach → teacher  
kind → kindness

6. Free and Bound Morphemes  
Free Morpheme: Can occur independently.  
Examples: book, happy, play, Bound Morpheme  
Cannot normally occur independently.  
Examples: un-, -ness, -ed, -s  
Example: Consider: "unhappiness"  
un- → prefix  
happy → root  
-ness → suffix  
Thus, morphological analysis breaks a complex word into meaningful components.

**Conclusion**  
The main components of words include roots, stems, prefixes, suffixes, inflectional morphemes and derivational morphemes. Understanding these components helps NLP systems analyze and process natural language.

3. What is Document Structure? Explain.

Introduction: A document is a collection of textual information organized in a meaningful form. Document structure refers to the way information is arranged within a document.

A document can contain different levels of organization, from the complete document down to individual words.

Levels of Document Structure

1. Document: The complete textual object is called a document.

Examples: Research paper, Newspaper article, E-mail, Web page, Report

2. Sections: A long document may be divided into sections.

Examples: Introduction, Methodology, Results, Conclusion

3. Paragraphs: Each section can contain multiple paragraphs.

A paragraph normally presents a particular idea or topic.

4. Sentences: Paragraphs are divided into sentences.

A sentence generally expresses a complete thought.

5. Words / Tokens: Sentences are divided into individual words or tokens.

For example: "Text mining is useful."

→ Text | mining | is | useful

Tokenization is an important step in text analysis.

6. Terms and Concepts: A term may be a single word or multiword phrase extracted from a corpus using NLP methods.

A concept represents a higher-level abstraction derived from a collection of documents.

Importance of Document Structure

Document structure helps in: Organizing information.

Understanding relationships between different parts.

Searching documents efficiently.

Extracting important information.

Classifying documents.

Summarizing large documents.

Building indexes.

Example: Consider a research paper:

Document

↓

Sections

↓

Paragraphs

↓

Sentences

↓

Words / Tokens

↓

Terms and Concepts

This hierarchical structure allows computers to process documents systematically.

Conclusion: Document structure provides an organized representation of textual information. Understanding the structure from documents to sections, paragraphs, sentences and tokens is important for NLP and text-mining applications.

4. Explain the Concept of Tokens and Lexemes

Token: A token is an individual unit obtained when text is divided during tokenization.

Consider: "Students study NLP."

The tokens are: Students, study, NLP

Tokenization is the process of splitting text into individual words or tokens.

Types of Tokens

Tokens may include:

Words

Numbers

Punctuation marks

Special symbols

Abbreviations

For example:

"Hello, world!"

Tokens can be:

Hello | , | world | !

Lexeme

A lexeme is an abstract vocabulary unit representing a group of related word forms.

For example:

GO is a lexeme that can have forms:

go

goes

went

going

These different forms represent the same basic lexical concept.

Token vs Lexeme

Token Lexeme

Actual occurrence in text Abstract vocabulary unit

Appears in a particular document Represents related word forms

Example: "went" Example: GO

Used during tokenization Used in lexical/morphological analysis

Example

Sentence: "The boys played games."

Tokens: The, boys, played, games

Possible lexemes: THE, BOY, PLAY, GAME

Thus, tokens represent the actual words appearing in the text, while lexemes represent the underlying vocabulary units.

Importance

Tokens and lexemes are useful for:

Text parsing

Information retrieval

Text classification

Morphological analysis

Search engines

Text mining

Conclusion: Tokens are the individual units found in text, while lexemes are abstract lexical units representing related word forms. Both are important for linguistic analysis and NLP.

6. Explain Parsing Using Deep Neural Networks

Introduction: Parsing is the process of analyzing a sentence to determine its grammatical or syntactic structure.

Traditional parsers use manually designed grammar rules or statistical models. Modern NLP increasingly uses deep neural networks to learn parsing patterns directly from data.

Basic Process

1. Input Sentence: The system receives a sentence.

Example: "The student reads a book."

2. Tokenization: The sentence is divided into tokens:

The | student | reads | a | book

3. Word Representation: Each word is converted into a numerical representation called an embedding.

Embeddings allow the model to capture relationships between words.

4. Neural Network Processing: A deep neural network processes the representations and learns relationships between words.

Possible architectures include: Recurrent Neural Networks, LSTMs, Transformers, Other deep learning architectures

5. Syntactic Structure: The model predicts relationships between words.

For example:

student → reads

book → reads

Thus, the system can construct a dependency or constituency structure.

Dependency Parsing

Dependency parsing identifies relationships between words.

Example:

Student → subject of → reads

Book → object of → reads

Constituency Parsing

Constituency parsing divides a sentence into hierarchical phrases.

Example: [The student] [reads [a book]]

The parser creates a tree representing the structure.

Advantages

Learns complex language patterns automatically.

Requires fewer manually written grammar rules.

Can handle large amounts of training data.

Captures contextual relationships.

Useful for complex NLP tasks.

Applications

Deep neural network parsing can support:

Question answering

Machine translation

Information extraction

Text summarization

Sentiment analysis

Natural language understanding

The uploaded NLP material emphasizes that parsing and language understanding are important for extracting concepts and relationships from text.

Conclusion

Deep neural network parsing uses learned representations and neural models to identify the syntactic structure and relationships within sentences. It is an important technique for modern NLP systems.

7. What is Ambiguity Resolution?

Introduction: Ambiguity occurs when a word, phrase or sentence has more than one possible interpretation.

Ambiguity resolution is the process of selecting the most appropriate interpretation using context and linguistic information.

Types of Ambiguity

1. Lexical Ambiguity: A word has multiple meanings.

Example: "Bank", Financial institution, Side of a river

The correct meaning depends on context.

2. Syntactic / Structural Ambiguity: A sentence can have more than one grammatical structure.

Example: "I saw the man with a telescope."

Possible meanings: I used a telescope to see the man., The man had a telescope.

3. Semantic Ambiguity: The words may have a clear structure but the overall meaning can have multiple interpretations.

4. Pragmatic Ambiguity: Meaning depends on the situation or context.

Ambiguity Resolution Methods

1. Context: Nearby words provide clues about the correct meaning.

2. Syntactic Analysis: The grammatical structure can help identify the intended interpretation.

3. Semantic Information: The meaning and relationships between words can be used.

4. Statistical Methods

Probabilities can be used to select the most likely interpretation.

5. Machine Learning

A model can learn ambiguity patterns from training examples.

Importance

Ambiguity resolution is important for:

Machine translation

Question answering

Speech recognition

Information extraction

Search engines

Chatbots

Conclusion

Ambiguity resolution enables computers to choose the most appropriate meaning or structure when multiple interpretations are possible. Context, syntax, semantics and statistical information are commonly used.

8. Explain Semantic Parsing, Structural Ambiguity and Predicate-Argument Structure

A. Semantic Parsing

Definition: Semantic parsing converts a natural-language sentence into a structured representation of its meaning.

It goes beyond simply identifying grammatical structure.

Example: Sentence: "John bought a car.", Semantic representation can identify: Action → buy, Buyer → John, Object → car

Applications: Question answering, Information extraction, Dialogue systems, Natural language understanding, Database query systems

B. Structural Ambiguity: Structural ambiguity occurs when a sentence can have more than one possible grammatical structure.

Example: "I saw the man with a telescope."

Interpretation 1: I used a telescope to see the man.

Interpretation 2: The man had a telescope.

Therefore, the same words can form different structures.

C. Predicate-Argument Structure: Predicate-argument structure represents an action, event or relationship and the participants involved in it.

5. Applications of Document Structure

Introduction: Understanding document structure allows NLP and text-mining systems to use not only individual words but also the relationships between sections, paragraphs and sentences.

Major Applications

1. Information Retrieval

Helps search engines locate relevant documents.

Important sections and terms can be indexed.

Users can retrieve relevant information more efficiently.

2. Document Classification

Documents can be classified according to their content.

Examples: Sports, Politics, Business, Technology

Text classification assigns documents to predefined categories.

3. Document Clustering

Similar documents can be grouped together.

No predefined categories are required.

Useful for organizing large document collections.

4. Text Summarization

Important sections and sentences can be identified.

Long documents can be reduced to shorter summaries.

This saves the reader's time.

5. Information Extraction

Important information can be extracted from documents.

Examples:

Names

Organizations

Places

Dates

Relationships

Information extraction and named-entity extraction are important text-mining applications.

6. Question Answering

Document structure helps locate relevant information.

Systems can search documents to find answers to user questions.

This is an important NLP application.

7. Academic Research

Document structure helps researchers:

Organize research papers.

Find relevant sections.

Identify research topics.

Discover relationships between concepts.

8. Web and Social Media Analysis

The structure of Web pages, posts and other documents can be analyzed to identify:

Topics

Trends

Opinions

Relationships

Conclusion: Document structure is useful in search, classification, clustering, summarization, information extraction, question answering and research. It enables computers to understand documents beyond simple word counting.

The predicate generally represents the action or relationship.

The arguments represent the entities participating in that action.

Example: "John ate an apple.", Predicate → ate, Argument 1 → John, Argument 2 → apple

So: EAT(John, Apple)

Another Example: "Mary gave John a book.", Predicate → gave, Giver → Mary, Receiver → John, Object → book

Representation: GIVE(Mary, John, Book)

Importance: Predicate-argument structures help NLP systems: Understand sentence meaning., Identify relationships between entities., Extract events., Answer questions., Perform semantic search., Support machine translation.

Conclusion: Semantic parsing identifies the meaning of a sentence, structural ambiguity deals with multiple possible grammatical structures, and predicate-argument structure represents actions and their participants. Together, they help computers move from simple text processing toward language understanding.

9. Detail About Models in Ambiguity Resolution Parsing

Natural language is ambiguous because the same word or sentence can have multiple interpretations. Parsing models help identify the most probable syntactic or semantic interpretation.

1. Rule-Based Parsing Model: Uses manually written grammar rules.

Applies rules to construct possible sentence structures.

The correct parse is selected according to the grammar.

Advantage: Easy to understand.

Limitation: Difficult to create rules for all natural-language cases.

2. Probabilistic Context-Free Grammar (PCFG): Extends grammar rules by assigning probabilities to them.

If multiple parses are possible, the model selects the parse with the highest probability.

For example: Sentence → NP + VP

may have a probability associated with it.

Advantage: Handles structural ambiguity using probabilities.

Limitation: Requires training data and may make simplifying independence assumptions.

3. Statistical Parsing: Uses probabilities learned from a treebank or parsed corpus.

Different possible parses are scored., The most probable parse is selected.

This is useful because language patterns can be learned from actual examples.

4. Dependency Parsing: Represents words as nodes and grammatical relationships as links.

Identifies relationships such as: Subject, Object, Modifier, Predicate,

Example: "John eats food.", John → subject, eats → predicate, food → object

5. Neural Parsing: Uses neural networks to learn syntactic patterns from data.

Word embeddings provide useful representations.

Modern systems may use deep neural networks and Transformer-based representations.

Neural models can capture complex contextual relationships.

Advantages: Learns complex patterns.

Reduces dependence on manually written rules.

Handles contextual information effectively.

6. Semantic-Based Resolution: Uses meaning to select the correct parse.

If two syntactic structures are possible, semantic information helps determine which one makes sense.

7. Context-Based Resolution: Uses surrounding words and sentences.

Helps resolve word-sense and structural ambiguity.

Particularly useful when a word has multiple meanings.

Example: "River bank" → bank means river side., "Bank account" → bank means financial institution.

Overall Process

Input Sentence → Tokenization → Possible Parses → Synthetic Analysis → Semantic + Contextual Information

→ Scoring/Probability → Best Parse Selected